Paper Plane Controller - Unity Asset Documentation

Overview

The Paper Plane Controller provides arcade flight physics, aerodynamics, and control mechanics for paper planes in Unity. It includes advanced lift and drag calculations, thrust control, ground detection, and visual enhancements such as mesh flutter and trail effects.

Features

- **Aerodynamics:** Lift, drag, and stall mechanics.
- **Thrust Control:** Adjustable forward thrust and speed with boost functionality.
- **Player Controls:** Yaw, pitch, roll, and stunt maneuvers.
- **Dynamic Environment Interactions:** Ground detection and air resistance.
- **Mesh Deformation:** Simulated paper flutter for enhanced realism.
- **Visual Effects:** Wing rotation, plane tilt, and trail renderers.

PaperPlanePhysics.cs

Purpose:

Handles aerodynamics, thrust, control input, and visual updates for the paper plane.

Key Components:

1. Aerodynamics

- `liftCoefficient` - Determines how much lift is generated.
- `dragCoefficient` - Controls air resistance.
- `stallAngle` - Angle at which the plane stalls and loses lift.
- `stallDragMultiplier` - Increases drag when stall occurs.

2. Thrust & Speed

- `baseThrust`- Base thrust applied to move forward.
- `boostMultiplier` - Increases thrust when boosting.
- `maxSpeed` - Limits the maximum velocity.

- `lateralThrust`- Applied side ways

3. Control Input

- `yawSpeed`, `pitchSpeed`, `rollSpeed` - Adjust turning rates.
- `maxPitchAngle` - Prevents excessive tilting.

4. Visuals & Effects

- Wing tilting and model rotation based on input.
- Trail renderers for left and right wings.

Functions:

- `GatherInputs()` - Captures player input.
- `ApplyAerodynamicForces()` - Computes lift and drag.
- `ApplyThrust()` - Applies thrust and limits velocity.
- `UpdateRotation()` - Adjusts rotation based on control input.
- `HandleGroundCheck()` - Detects if the plane is grounded.
- `HandleReset()` - Resets plane position if stuck.

PaperPlaneMeshMod.cs

Purpose:

Applies real-time mesh deformation to simulate paper flutter.

Key Components:

- `flutterIntensity`, `flutterSpeed` - Controls flutter effect.
- `inputFlutterMultiplier` - Increases flutter with control input.

Functions:

- `AdjustFlutterIntensity()` - Modifies flutter based on speed and input.
- `ApplyFlutterEffect()` - Animates vertex positions.

How to Use

1. **Setup**:
    - Attach `PaperPlanePhysics` to your plane GameObject.
    - Assign necessary transforms (model, wings, trail renderers).
    - Adjust settings for flight behavior.
    - Or Simply drag PaperPlane Controller prefab to the scene
2. **Controls**:
    - `W/S` - Pitch up/down.
    - `A/D` - Yaw left/right.
    - `Q/E` - Roll left/right.
    - `Shift` - Boost.
    - `R` - Reset position.
3. **Customization**:
    - Modify lift, drag, and thrust values for different flight styles.
    - Adjust mesh flutter parameters for varying effects.

Notes

- Ensure the Rigidbody has gravity enabled.
- Adjust drag and lift values for optimal flight behavior.
- Use smooth interpolation for better visual effects.

Contact & Support

For issues or feature requests, contact us at:
📧 **ragerungames@gmail.com**
🌐 **https://discord.gg/ZHxB9MkkR6**

Enjoy flying with the **Paper Plane Controller**! ✈️